

OMAR AHMED ELSAYED

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Professional Summary

Data Engineer II with production experience in Snowflake, dbt, PostgreSQL, and LLM integration. Reduced dbt build times by **70%**, architected pipelines processing **hundreds of GBs** weekly, and shipped an SQL-based RAG Natural Language-to-SQL application. Active open-source contributor (Wren AI) and Arab AI & IoT Challenge winner.

Technical Skills

Data Engineering: dbt, Snowflake, PostgreSQL, Jinja, ELT/ETL, Data Modeling, Blob Storage, Web Scraping
AI & LLMs: LLM Ops, RAG, Vector Databases, Wren AI (Text-to-SQL), Agentic AI
Languages: Python, SQL, Bash, Markdown | Exposure: Go, C/C++, C#, JavaScript, VHDL, Dart, LaTeX
Cloud & DevOps: Azure DevOps, AWS, Docker, Linux, Git, CI/CD, Snowpark, Managed Identity
Agentic AI & Productivity: Claude Code, Codex, OpenCode, GitHub Copilot
Visualization: Power BI, Streamlit, React (web), Plotly, Matplotlib, Seaborn

Education & Certifications

Arab Academy for Science, Technology and Maritime Transport (AAST)

B.Sc. Computer Engineering

Certifications:

- **ALX 13-Month Data Science** (ExploreAI - sponsored by Mastercard) – [Certificate](#)
- **NVIDIA DLI:** Fundamentals of Deep Learning – [Certificate](#)
- **Udacity:** Machine Learning Cross-Skilling Nanodegree – [Certificate](#)

Professional Experience

Data Engineer II

Apr 2026 – Present

Raisa Energy

Cairo, Egypt

- **Upstream Feed Migration:** Migrated a critical third-party upstream feed from legacy ETL middleware to a direct Snowflake feed; validated **5.2M+ wells** and **599 columns** with 100% match, 0.00% volume variance, zero breaking changes, and **\$30K/year** in avoided licensing/compute costs
- **Accounting Pipeline Modernization:** Rewrote a full-table accounting export as an incremental shadow-table pipeline; used row-hash change tracking to exact-match validate up to **42M rows** and scope failures to affected business groups only
- **Snowpark Containerization:** Containerized a third-party Python extraction pipeline for Snowpark with OAuth2, external access, persistent log volumes, and scheduled execution
- **Cross-Functional Coordination:** Aligned accounting teams and third-party providers on stable source identifiers, effective-date rules, and publish semantics for downstream integrations
- **Agentic Development:** Used Claude Code, Codex, OpenCode, and GitHub Copilot to accelerate Python pipeline development, debugging, code review, and documentation

Data Engineer I

Oct 2024 – Apr 2026

Raisa Energy

Cairo, Egypt

- Engineered dbt incremental strategies and custom SCD Type II materializations, cutting pipeline build times by **70%**
- Built extraction pipelines for APIs, websites, and emails; ingest **100+ GB** of unstructured data weekly into Blob Storage and Snowflake
- Developed an SQL-based RAG Natural Language-to-SQL app, enabling non-technical stakeholders to self-serve analytics
- Architected Azure DevOps CI/CD with automated testing, key-pair auth (case study)
- Automated GCM authentication with Azure UAMI, eliminating manual PAT rotation and silent auth failures in pipelines
- Authored dbt/SQL best practices; built an issue-tracking integration linking Azure DevOps, Slack, and email

Data Engineering Intern

Jul 2024 – Sep 2024

Raisa Energy

Cairo, Egypt

- Modernized legacy infrastructure and optimized data models powering mission-critical Power BI dashboards

Open Source & Projects

Open Source: Wren AI (Text-to-SQL) | Contributor

- Resolved critical backend bugs and merged PRs with the Wren AI core engineering team

AIoT Smart Shopping Cart | Data Visualization & Frontend Engineer

- Winner of Arab AIOT Challenge & EGAIOT; 2nd Place AUC Hackathon; exhibited at GITEX Global
- Partnered with Spinneys and Quanttronix; built real-time Analytics Dashboard and Expo (React Native) consumer app

Intelligent Maze Agent (Dell Hacktrick) | Machine Learning Engineer

Mar 2023

- Ranked top 10 of 200+ teams; developed a DQN agent with TensorFlow/PyTorch on AWS EC2

Volunteering & Leadership

Technical Instructor & Lecturer

Sep 2022 – Sep 2024

IEEE AAST Cairo Student Branch

Cairo, Egypt

- Designed Python workshops and mentored 100+ students in core coding and debugging; led technical R&D sessions